



2024 Year in Review

Internet patterns and trends observed through
Cloudflare's global network during 2024

HIGHLIGHTS

17% Global Traffic
Growth

3.3x Starlink
Surge

225 Major
Disruptions

6.5% Traffic
Mitigated

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic continues its upward trajectory, with satellite connectivity emerging as a transformative growth driver



Global Traffic Growth YoY

17% ▲ continuing upward trend



Starlink Traffic Surge

3.3x ▲ satellite expansion



Mobile Device Traffic

41% of global traffic



iOS Traffic Share

33% of all traffic

Normalized Starlink Traffic Worldwide — 2024



Google Leads Top 10 Services as Generative AI Rises

Google holds #1 for the second consecutive year; generative AI platforms continued rapid ascent while the Metaverse faded

RANK	SERVICE	CATEGORY
#1	Google	Search / Platform
#2	Facebook	Social
#3	Apple	Technology
#4	TikTok	Social / Video
#5	AWS	Cloud
#6	Microsoft	Technology
#7	Instagram	Social / Photo
#8	YouTube	Video
#9	Amazon	E-commerce
#10	WhatsApp	Messaging

Based on Cloudflare 1.1.1.1 DNS resolver data



Generative AI: The Defining Trend of 2024

Generative AI platforms continued their rapid rise throughout 2024, reshaping how users interact with Internet services for search, content creation, and productivity.

▲ Generative AI Rising

▼ Metaverse Fading



Search Engine Referral Traffic Share



Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Chrome 124's default enablement in April triggered a 6.5x surge in post-quantum adoption; HTTP/3 and Go lead protocol and language shifts



Post-Quantum Encryption Adoption

~2% → ~13% of TLS 1.3 traffic after Chrome 124 (Apr 17)
Firefox rolling out default PQ | Safari initial testing



HTTP/3 Adoption

21% of traffic uses HTTP/3 (QUIC-based)

Faster connections, packet-loss mitigation, encryption by default



Top API Client Languages

Go	~12%
Node.js	~10%
Python	9.6%
Java	7.4%
.NET	3.6%

Browser Market

Chrome

Most popular browser globally

Google dominates search (89%)

Mobile OS Share

33% iOS

Peak Android: >95% some regions

Peak iOS: ~66% some regions

IPv6 Adoption

28% of dual-stack traffic

Top Download Speed

Spain highest among all countries

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Government-directed shutdowns accounted for over half of all major Internet disruptions, underscoring connectivity fragility

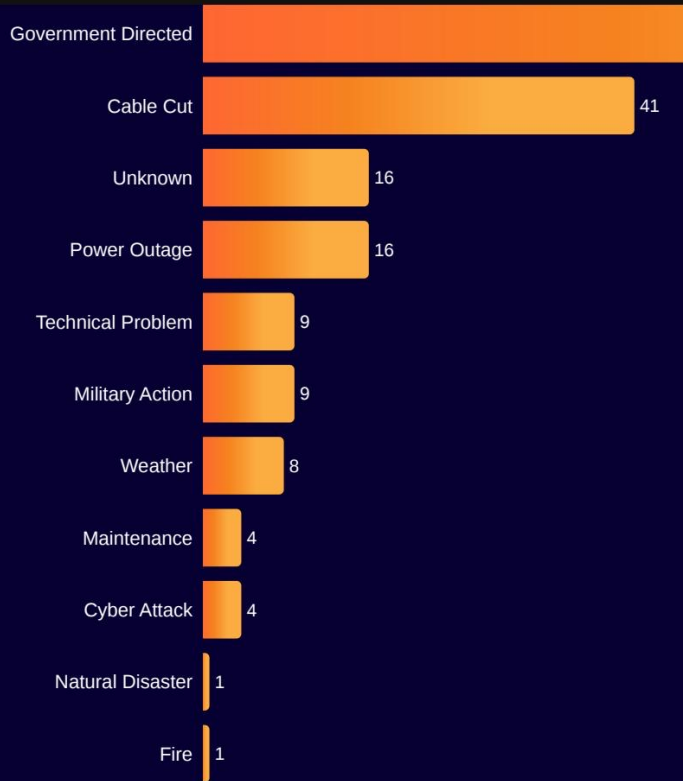
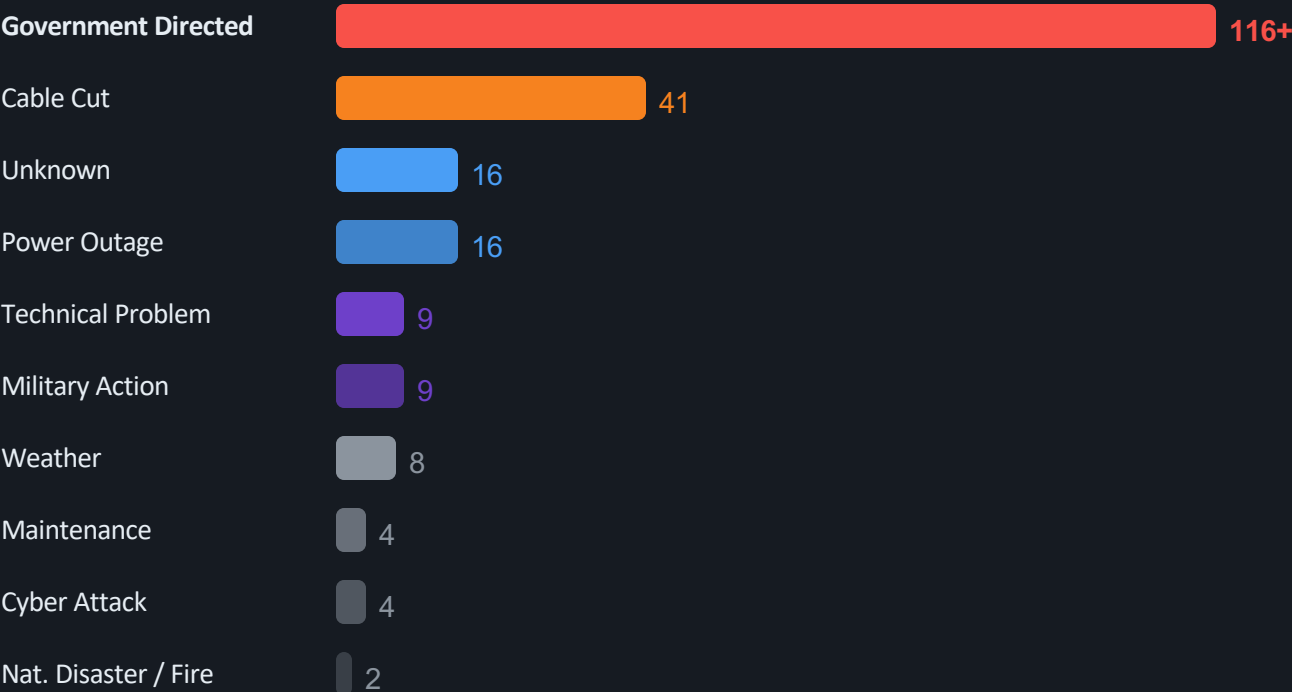
 Total Major Disruptions

225 observed globally

 **Government Shutdowns Dominate**

116+ government-directed shutdowns account for over 51% of all observed disruptions in 2024

Internet Disruption Causes — 2024



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security threats remain pervasive with Log4j still exploited 2+ years post-disclosure; email-borne threats continue evolving



Traffic Mitigated

6.5% of all traffic
processed by Cloudflare



Most Attacked Sector

**Gambling
& Games**



Bot Traffic from Top 10 Countries

68% of global bots
high geographic concentration



RPKI Valid Routes Increase

+6.4%
improved routing security



Log4j: Persistent Exploitation

CVE-2021-44228

First disclosed: December 2021

Still actively targeted by attackers more than two years later,
underscoring the persistent nature of widely known vulnerabilities.

CRITICAL — Active Exploitation



Email Security Threats

4.3% of emails were malicious

43% contained deceptive links

Top malicious/spam TLD: **.bar**

Threat Landscape Summary

Attack Volume

6.5% of all traffic mitigated
Gambling/Games most targeted

Bot Concentration

68% from just 10 countries
High geographic clustering

Routing Security

RPKI valid routes +6.4%
Positive step for Internet integrity

Five Takeaways Shaping the 2025 Internet Outlook

Strategic findings from Cloudflare's global network data to inform infrastructure and security planning for 2025

01 Traffic Growth Trajectory

Global Internet traffic grew 17% YoY. SpaceX Starlink surged 3.3x, bringing satellite connectivity to underserved regions and expanding the addressable Internet population.

02 Post-Quantum Encryption Momentum

Adoption leapt from ~2% to ~13% of TLS 1.3 traffic after Chrome 124 default enablement in April 2024. Firefox and Safari following suit signals industry-wide crypto agility transition.

03 Connectivity Fragility from Government Actions

Government-directed shutdowns led 225 major Internet disruptions (116+ incidents, >51% of total). Cable cuts (41) and power outages (16) round out the top three causes.

04 Persistent Security Threats

6.5% of all traffic mitigated. Log4j exploitation persists 2+ years post-disclosure. Gambling/Games most targeted sector. 4.3% of emails malicious, 43% containing deceptive links.

05 Evolving Developer Ecosystem

Go became the top API client language (~12% of automated API traffic). HTTP/3 reached 21% adoption. Generative AI platforms continued rapid rise, reshaping how users interact with Internet services.